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Research paper

Day-ahead scheduling of microgrid with hydrogen energy considering economic and environmental objectives

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ABSTRACT

Conventional microgrids face challenges such as high costs, waste heat, and significant polluting emissions. Hydrogen energy systems can reduce costs and produce zero emissions. Therefore, this paper proposes a co-scheduling method for power and equipment waste heat to address the economic and environmental dispatch (EED) problem of hydrogen microgrids. Firstly, this paper establishes a co-scheduling model for power and heat in hydrogen microgrids without thermal storage. Then, in terms of optimization methods, this study introduces the Improved Honey Badger Algorithm (IHBA), which incorporates piecewise mapping initialization and a segmented optimal decreasing Levy flight strategy. Sensitivity analysis is employed to optimize the parameters of the IHBA and to compare its performance with other methods. Additionally, this paper presents a comparison framework based on the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) model to identify algorithms that achieve an optimal balance of solution quality and efficiency for the EED problem in hydrogen microgrids. The combined cost of the hydrogen microgrid is reduced by 16.47 % compared to a conventional microgrid. Moreover, the IHBA algorithm reduces the combined cost of the hydrogen microgrid by 5.5 % compared to the standard HBA algorithm.

1. Introduction

According to forecasts, global energy demand is projected to increase by 43 % by the year 2040. In certain regions, power generation accounts for over 2/5 of air pollutants such as SO_2 and NO_X (Chen et al., 2020; Wang et al., 2023a). Therefore, the immediate priority is to optimize the use of clean energy to ensure the reduction of energy costs and alleviate environmental pollution. Converting renewable energy into hydrogen energy is an ideal utilization method (Ghenai et al., 2020). The cost of hydrogen production and utilization is relatively high. As microgrids continue to develop, they are becoming a feasible solution to overcome the challenges hindering the application of hydrogen. Within the microgrid, hydrogen production and utilization can occur concurrently, with the additional capability of converting electricity into hydrogen for storage purposes. Microgrids facilitate the incorporation of different clean energy sources to cater to the simultaneous demands of electrical, heat, and cooling loads (Fonseca et al., 2019). In this context, hydrogen energy is crucial for generating power and consuming excess energy to meet electricity load demands. The heat energy generated in this process enhances energy utilization to meet the heat demands and avoids polluting gas production, rendering hydrogen an ideal resources for microgrids broadening the application scenarios for hydrogen energy (Vivas et al., 2018). However, integrating the production and utilization of hydrogen energy into microgrids is a complex challenge, necessitating transforming the original power and heat supply structure. Additionally, this introduces further dimensions to the dispatching problems associated with microgrids.

In the study of microgrids containing hydrogen energy, Fang et al (Ruiming, 2019). employs an enhanced NSGA-II methodology to optimize an integrated energy system with electrolytic hydrogen, hydrogen storage tanks, and fuel cell units, but it only explores the feasibility of fully adopting hydrogen energy storage for power scheduling. Ju et al (Ju et al., 2023). models an island microgrid containing hydrogen production units and fuel cells, optimizing it using a hybrid differential evolution technique. Although waste heat from a gas turbine was utilized, the study did not consider the utilization of waste heat from the hydrogen production unit and fuel cells. Mohseni et al (Mohseni and Brent, 2020). evaluates microgrid systems using hydrogen as an energy storage medium and analyzes the efficacy of 20 meta-heuristic algorithms in addressing optimization design challenges. Wang et al (Wang et al., 2023a). optimizes the scheduling of an integrated energy system

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Nomenclature

ADP	Approximate dynamic programming.
EES	Energy Storage Systems.
EED	Economic Environment Dispatching.
E2H	Electric-to-thermal.
ELE	Electrolyzer.
FC	Fuel cells.
GWO	Gray wolf optimization algorithms.
HBA	Honey Badger Algorithm.
HESS	Hydrogen energy storage system.
HT	Hydrogen storage tank.
HHO	Harris Hawks Optimization Algorithm.
IHBA	Improved Honey Badger Algorithm.
MILP	Mixed-integer linear programming.
MG	Microgrid.
MT	Micro turbine.
MRFO	Manta Ray Foraging Optimization Algorithm.
PSO	Particle swarm optimization algorithms.
PV	Photovoltaic.
RIME	RIME Algorithm.
TOPSIS	Technique for Order Preference by Similarity to the Ideal Solution.
WT	Wind turbine.

that includes a hydrogen energy storage system, using the waste heat from the hydrogen energy storage system. However, it does not consider the energy loss of the waste heat boiler. Some existing studies on microgrid scheduling fail to consider hydrogen energy utilization and thermal scheduling. Moreover, research that does consider hydrogen energy often overlooks the utilization of waste heat from hydrogen systems and the losses associated with thermal energy storage.

Currently, the methods for scheduling optimization of microgrids primarily include mathematical programming methods and meta-heuristic algorithms. Many researchers have employed Approximate Dynamic Programming (ADP) (Shuai et al., 2019) and Mixed Integer Linear Programming (MILP) techniques (Mirhoseini and Ghaffarzadeh, 2020) for microgrid scheduling optimization. Despite the effectiveness of mathematical programming methods, they often face increased runtime and limitations in terms of the consistency and differentiability of the goal function, making them less adept at tracking dynamic changes. In contrast, meta-heuristic algorithms address some challenges, making them widely applicable in microgrid design and optimization. Traditional meta-heuristic algorithms, such as genetic algorithms (Roy et al., 2020), particle swarm optimization algorithms (PSO) (Nikmehr and Ravadanegh, 2015), and gray wolf optimization algorithms (GWO) (Sharma et al., 2016), have been used by researchers to reduce operating costs or adjust load requirements. Other researchers use the newly proposed meta-heuristic algorithm for optimization. Zheng et al (Zheng et al., 2023). employed the Chaotic particle swarm algorithm to examine the influence of various energy storage techniques on the benefits aspects of the small scale energy system. Li et al (Li et al., 2021). implemented a technique called the thermal storage tank state-based operating strategy, utilizing the multi-objective seagull algorithm to obtain optimization result of economic, energy, and pollution objectives. However, the No Free Lunch hypothesis believes that no individual meta-heuristic approach can universally offer the optimal answer for all optimization issues. Because the process is stochastic, other algorithms may discover superior answers inside the search space (W.G.M. D.H. Wolpert, 1997). Hence, it is beneficial for science to construct novel meta-heuristic algorithms or to increase the functionality of existing algorithms, which may improve the effectiveness of optimization accuracy for various problems. V et al (K. V et al., 2022). introduced the Mimosa pudica algorithm as a solution for microgrid energy management. The program's main objective is to reduce the total production cost by resolving the power flow issue. Liu et al (Liu et al., 2024). presented an improved chameleon algorithm based on strategies such as tent chaotic mapping, dynamic weight factor, and Multi-strategy prey search. This approach verified the cost of the integrated cooling, heating, and electricity microgrid system in many typical seasonal conditions. Some research on microgrid scheduling optimization algorithms often overlooks improving key parameters of these algorithms to enhance performance. Meanwhile, these studies compare the performance of the proposed improved algorithms with the original algorithm and other algorithms based on the optimization objective, but they only use the optimization results for comparison. However, search efficiency is another important factor, and search time should be included in the evaluation. A balanced consideration of search capability and efficiency is required to identify suitable algorithms for scheduling optimization of microgrids containing hydrogen energy.

Based on the above analysis, we propose a novel microgrid model integrating hydrogen energy to efficiently manage waste heat in real-time while fulfilling heat demands. Furthermore, we enhance the efficiency of optimization and cost reduction by refining the HBA algorithm, utilizing sensitivity analysis to tailor algorithm parameters to the optimization task. Lastly, we propose a TOPSIS-based algorithm comparison framework to select the most suitable algorithm based on optimization search capability and efficiency. This study offers valuable insights for addressing day-ahead scheduling challenges in hydrogen energy microgrids.

The rest of this study is organized as follows: Chapter 2 examines the foundational structure of a microgrid incorporating hydrogen energy storage and formulates a mathematical model for microgrid units. Chapter 3 provides a comprehensive explanation of the EED model. Chapter 4 outlines the enhancements made to the HBA algorithm. Chapter 5 offers a simulation and analytical investigation of a fishery farm with electrical and thermal energy requirements in southern China. Finally, Chapter 6 concludes the paper and proposes avenues for future research.

2. Methodology

In a microgrid that incorporates hydrogen, energy inputs originate from renewable sources, hydrogen, and natural gas, while outputs include electricity, heat, and hydrogen. Electricity is generated by photovoltaic (PV), wind turbine (WT), fuel cell (FC), microturbine (MT), energy storage system (ESS), and the main grid. Thermal energy is produced by FC, electrolyzer (ELE), MT, and electric-to-thermal (E2H) units. The hydrogen energy storage system (HESS) consists of ELE, a hydrogen storage tank (HT), and FC, with hydrogen being either purchased from a gas station or sold. The ELE generates hydrogen and stores it in the HT when there is excess power or during periods of low electricity prices. The stored hydrogen is then used by the fuel cell to produce electrical and thermal energy during peak electricity price periods and high thermal energy demand. The battery storage system is recharged when electricity prices are low and there is a surplus of electricity, and it discharges to supply electricity when prices are high or the supply is low. The MT consumes natural gas to produce both electricity and heat. The E2H is used to produce heat from electricity when there is a shortage of thermal energy. Both electrical and thermal energy are supplied to meet the system load requirements. Fig. 1 provides a detailed depiction of the generation, conversion, transmission, and storage processes of the various energy flows within the microgrid. In this figure, the red lines represent the thermal network, the blue lines represent the electrical network, and the green lines represent the

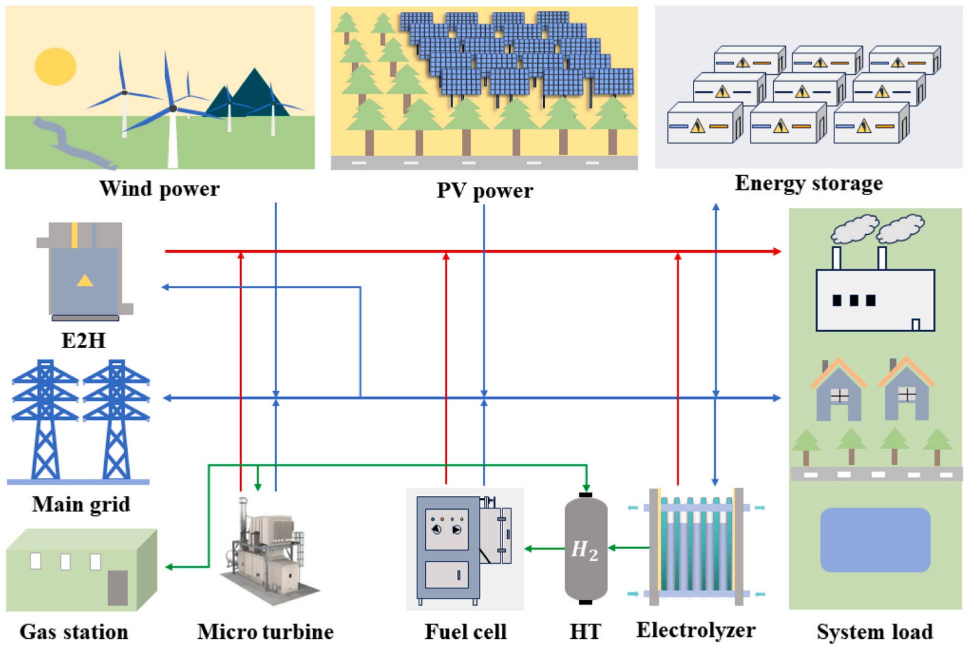

Fig. 1. Integrated energy framework for electric-hydrogen hybrid microgrid system.

hydrogen and natural gas networks.

2.1. Electricity-hydrogen hybrid microgrid system model

2.1.1. PV model

Photovoltaics convert light radiation into electrical energy. The primary elements that influence the generation of power from photovoltaic systems are the intensity of light radiation and the temperature, and the output power $P_{pv}(t)$ is defined as (Zhang et al., 2023):

$$P_{pv}(t) = P_{STC} \cdot G / G_{STC} \cdot [1 + k_p (T_c - T_{STC})] \tag{1}$$

Where P_{STC} is the rated power ($G_{STC} = 1000W/m^2$, $T_{STC} = 25°C$), G, k_p and T_c are the light radiation intensity, power temperature coefficient, and environmental temperature, respectively.

2.1.2. WT model

The WT use wind power to generate electricity. The output power $P_{WT}(t)$ of wind turbine can be expressed as (Emrani et al., 2022):

$$P_{WT}(t) = \begin{cases} 0 & \nu < \nu_{in} \\ P_{rate} \times (\nu^3 - \nu_{in}^3)/(\nu_{rate}^3 - \nu_{in}^3) & \nu_{in} \le \nu \le \nu_{rate} \\ P_{rate} & \nu_{rate} < \nu < \nu_{out} \\ 0 & \nu > \nu_{out} \end{cases} \tag{2}$$

Where P_{rate} is the rated output power, ν_{in}, ν_{out} and ν_{rate} are the cut-in speed, cut-out speed, and rated wind speed, respectively. And ν is the wind speed.

2.1.3. MT model

Eqs. (3) and (4) describe the natural gas consumption and the heat generation during the operation of MT (Wang et al., 2023b):

$$P_{MT}(t) = G_{MT}(t)\eta_{MT\_E} \tag{3}$$

$$Q_{MT}(t) = G_{MT}(t)\eta_{MT\_Q} \tag{4}$$

Where $P_{MT}(t)$ is the output power of MT, $G_{MT}(t)$ is the natural gas power of MT, η_{MT_E} represents the electricity production efficiency. $Q_{MT}(t)$ is the thermal power output of MT, η_{MT_Q} is the heat production coefficient.

2.1.4. HESS model

Unlike devices in conventional microgrids, the HESS can generate both electrical and thermal energy. Additionally, it can store and consume electrical energy, and it is environmentally friendly. Electric-heat-hydrogen coupling characteristics of HESS is as follows (Ghenai et al., 2020; Wang et al., 2023a):

$$P_{ELE}(t) = H_{ELE}(t) q_{H2} / \eta_{ELE\_H} \tag{5}$$

$$Q_{ELE}(t) = P_{ELE}(t)\eta_{ELE\_Q}(1 - \eta_{ELE\_H}) \tag{6}$$

$$P_{FC}(t) = H_{FC}(t)\eta_{FC\_E} \tag{7}$$

$$Q_{FC}(t) = P_{FC}(t)\eta_{FC\_Q}(1 - \eta_{FC\_E}) \tag{8}$$

Where $P_{ELE}(t)$, $Q_{ELE}(t)$, $H_{ELE}(t)$ are the electricity consumption power, heat power and hydrogen production power of ELE; $P_{FC}(t)$, $Q_{FC}(t)$,$H_{FC}(t)$ are the electricity production power, heat power and hydrogen consumption power of FC; η_{ELE_H} and η_{ELE_Q} are the efficiency of hydrogen production and heat conversion in ELE; η_{FC_E} and η_{FC_Q} are the efficiency of electricity production and heat conversion in FC. The storage state $SOH(t)$ of HT is shown as below:

$$SOH(t) = V_{H_2}(t) / V_{H_2\ \max} \tag{9}$$

Where $V_{H_2}(t)$ and $V_{H_2\ \max}$ are the total hydrogen power at time t and maximum hydrogen energy capacity.

2.1.5. E2H model

The E2H unit converts electrical energy into thermal energy and has a simple structure. The thermal energy $Q_{E2H}(t)$ of E2H can be described as:

$$Q_{E2H}(t) = P_{E2H}(t)\eta_{E2H\_Q} \tag{10}$$

Where $P_{E2H}(t)$ is the electricity consumption power and η_{E2H_Q} is the heat conversion coefficient of E2H.

2.1.6. ESS model

By controlling the charge and discharge in a reasonable manner, the ESS could make the microgrid system more stable and economical. The state of charge SOC of the energy storage system is as follows (Zhou and Ai, 2019):

$$SOC_{t+1} = SOC_t(1-\delta) + P_{charge,t}\eta_c \Delta t / E_s - P_{d,t}\Delta t / \eta_d E_s \tag{11}$$

Where SOC_t is the state of charge at the time step t, δ represents the self-discharge rate, $P_{c,t}$ and $P_{d,t}$ are charging and discharging power of the battery at the period t, respectively, η_c and η_d are the efficiency of charging and discharging, respectively, E_s is the battery capacity.

2.2. Identification of best-performance algorithm using TOPSIS

The output of meta-heuristic algorithms can be used to construct an evaluation matrix (EM_{ij}), where i represents the number of algorithms and j represents the number of objectives.EM_{ij} undergoes a process of standardization and normalization to obtain a dimensionless matrix Em_{ij} as defined in Eq. (12), and $w_j$represents the weightage applied to each criterion (Sohani et al., 2022).

$$Em_{ij} = w_j EM_{ij} \Bigg/ \left(\sum_{i}^{I} (EM_{ij})^2 \right) \tag{12}$$

Then, the optimal and non-optimal solutions are defined as Eq. (13), where d_i^+ and d_i^- represent the ideal solution and nonideal solution respectively.

$$\begin{cases} d_i^+ = \sqrt{\sum_{j=1}^{J} \left(Em_{ij} - r_j^+ \right)^2} \\ d_i^- = \sqrt{\sum_{j=1}^{J} \left(Em_{ij} - r_j^- \right)^2} \end{cases} \tag{13}$$

where r_j^+ represents the optimal point and r_j^- represents the non-optimal point. Ultimately, the approximation degree index C_i for each algorithm is as follows:

$$C_i = d_i^+ / (d_i^+ + d_i^-) \tag{14}$$

where C_i denotes the closeness degree between the solution and the ideal point. The method that offers the highest value for C_i is considered the optimal approach for solving the optimization issue.

3. Economic emission dispatch model

3.1. Objective function of economic emission dispatch problem

The purpose of the EED problem is to minimize the cost and pollution emissions of the microgrid system during the dispatching cycle. The objective function is specified as follows:

$$\min F = wF_1 + (1-w)F_2 \tag{15}$$

Where F is the objective function, F_1 is the cost of the microgrid system, F_2 is the cost of pollutant remediation. w is the scale factor that balances the cost of the microgrid system and the emission of pollutants.

$$F_1 = \sum_{t=1}^{T} \left(\begin{array}{l} c_{pv}P_{PV}(t) + c_{WT}P_{WT}(t) + c_{MT}P_{MT}(t) + c_{FC}P_{FC}(t) \\ + c_{ELE}P_{ELE}(t) + c_{HT}V_{HT}(t) + c_{ESS}P_{ESS}(t) \end{array} \right) + F_{gas} + F_{Grid} \tag{16}$$

Where c_{PV}, c_{WT}, c_{MT}, c_{FC}, c_{ELE}, c_{HT} and c_{ESS} are the operation and maintenance costs for PV, WT, MT, FC, ELE, HT and ESS. F_{gas} and F_{grid} represent the cost of natural gas and the cost of power exchange between the microgrid and the main grid, respectively. The cost of the E2H unit is ignored because they are extremely low.

$$F_{gas} = \sum_{t=1}^{T} \left(c_{gas}G_{MT}(t) + c_{H_2}(H_{ELE}(t) - V_{H_2\ \max}) \right) \tag{17}$$

Where c_{gas} is the cost factor of natural gas. c_{H_2} is the cost factor of H_2.

$$F_{Grid} = \sum_{t=1}^{T} \left[k_{buy}(t)P_{buy}(t) - k_{sell}(t)P_{sell}(t) \right] \tag{18}$$

where k_{buy} and k_{sell} are the price of electricity purchased and sold. $P_{buy}(t)$ and $P_{sell}(t)$ are power purchased and sold from the main grid. And the emission in microgrids primarily originate from the MT and the main grid.

$$F_2 = \sum_{t=1}^{T} \sum_{i=1}^{N} \sum_{q=1}^{q} \left[(c_q x_{iq}) P_i(t) + (c_q x_{gridq}) P_{buy}(t) \right] \tag{19}$$

Where q is the type of pollutant, including CO, CO_2, SO_2 and NO_x, c_q is the cost of qth pollutant. x_{iq} and x_{gridq} are the emission coefficients. And the parameter values are shown in the Appendix Table.

3.2. Constraint condition

(1) At the system level, the sum of outpower from the source and the load demand must meet the power balance requirements (Wang et al., 2022). Furthermore, given the short distances between components within the microgrid system and the length of the transmission lines, certain electrical characteristics and power losses are neglected to simplify the calculations under ideal conditions. Power balance constraint can be expressed as:

$$\begin{array}{l} P_{PV}(t) + P_{WT}(t) + P_{MT}(t) + P_{FC}(t) + P_{ESS}(t) \\ + P_{Grid}(t) - P_{ELE(t)} = P_{Load}(t) + P_{E2H}(t) \end{array} \tag{20}$$

Where $P_{ESS}(t)$ is the power of ESS, and P_{Grid} is the interaction power between the system and the grid, $P_{load}(t)$ is the load demand of the system.

(2) The constraint of thermal balance

$$Q_{MT}(t) + Q_{FC}(t) + Q_{ELE}(t) + Q_{E2H}(t) = Q_{Load}(t) \tag{21}$$

Where $Q_{Load}(t)$ is the heat load demand of the system.

(3) Generation power limit of micro-source

$$P_{i,\ \min}(t) \le P_i(t) \le P_{i,\ \max}(t) \tag{22}$$

Where $P_{i,\ \min}(t)$ is the minimum power of the ith unit, and $P_{i,\ \max}$ is the maximum power of the ith unit.

(4) Constraints on the power of interaction between the microgrid and the main grid

$$P_{Grid,\ \min} \le P_{Grid}(t) \le P_{Grid,\ \max} \tag{23}$$

Where $P_{Grid,\ \min}$ and $P_{Grid,\ \max}$ are the minimum and maximum grid interaction power.

(5) Constraints of energy storage

$$\begin{cases} P_{Bat,\ \min} \le P_{Bat}(t) \le P_{Bat,\ \max} \\ SOC_{\min} \le SOC(t) \le SOC_{\max} \end{cases} \tag{24}$$

$P_{Bat,\ \min}$ is the upper limit of the charging power of the battery, $P_{Bat,\ \max}$ is the upper limit of the discharge power of the battery, $SOC_{\min}$ and $SOC_{\max}$ are the upper and lower limits of the energy storage capacity, respectively.

4. Honey Badger algorithm and its improved algorithm

4.1. HBA algorithm

The HBA method is a distinctive meta-heuristic method that draws inspiration from honey badger hunting (Hashim et al., 2022). The predation process of honey badgers consists of two distinct phases: the digging phase and the honey phase. During the digging phase, the honey badger initially utilizes its olfactory ability to locate the approximate location of quarry and subsequently digs to capture the quarry. During the honey phase, the honey guide bird locates the beehive and directs the honey badger towards it. Then the hive is opened by honey badgers, and they participate in enjoying the reward of teamwork.

4.1.1. Initialization process

The following equation defines the initialization of honey badger population's location

$$x_i = lb_i + (ub_i - lb_i) \times r_1 \tag{25}$$

Where x_i is the position of i_{th} honey badger , ub_i and lb_i are upper bounds and lower bounds in the tracking area.r_1 is a random number between $[0, 1]$.

4.1.2. Definition of the intensity factor I and density factor α

A link exists between the proximity of the honey badger to its prey and the level of intensity. The velocity of the honey badger's movement is contingent upon the level of intensity; a higher intensity results in a swifter approach towards its prey, while a lower intensity leads to a slower approach. Intensity factor I_i is defined as

$$\begin{cases} I_i = r_2 \times M/\left(4\pi d_i^2\right) \\ M = (x_i - x_{i+1})^2 \\ d_i = x_{prey} - x_i \end{cases} \tag{26}$$

where$M$ is the source strength, d_i signifies the length between the quarry and the i_{th} honey badger, r_2 is an arbitrary value between $[0, 1]$, x_{prey} is the location of prey.

The density factor undergoes a nonlinear reduction over successive iterations, facilitating a smooth transition from exploratory behavior in the initial stages to exploitative tendencies in the later phases. The density factor can be described as follows:

$$\alpha = P \times \exp(-t/t_{\max}) \tag{27}$$

where P is a parameter that defaults to 2; the value of the present iterations is denoted by t; the max iterations is denoted by $t_{\max}$.

4.1.3. Updating population position

$k$is the random decision coefficient in the range between 0 and 1. When $k < 0.5$, honey badger updates the position of the digging mode; when$k \geq 0.5$, honey badger updates the position of the honey mode. The population's position update equation can be represented as:

$$x_{new} = \begin{cases} x_{prey} + F \times \beta \times I \times x_{prey} + F \times \alpha \times r_3 & k < 0.5 \\ \times d_i \times |\cos(2pr_4)| \times [1 - \cos(2\pi r_5)] & \\ x_{prey} + F \times r_7 \times \alpha \times d_i \quad k \geq 0.5 & \end{cases} \tag{28}$$

where x_{new} is the next position of the individual; x_{prey} is the global best location found so far; r_3, r_4, r_5 and r_7 are random numbers, which are in the range of $[0, 1]$; β is a constant that quantifies the honey badger's capacity to get honey. F is used to change the search direction and expand the search range according to the following formula:

$$F = \begin{cases} 1 & r_6 \leq 0.5 \\ -1 & r_6 > 0.5 \end{cases} \tag{29}$$

Where r_6 is a randomly generated number within the range of $[0, 1]$.

4.2. Improved honey badger algorithm

4.2.1. Piecewise map

The HBA algorithm is the same as most meta-heuristic algorithms. The random number approach is used to create the initial population. However, there is a certain degree of chance in the initial population of individuals created by the random number approach. The initial individuals are not uniformly distributed in the search area, and falling into the local optimum is easy. Chaotic mapping can effectively solve the random number method's shortcomings in generating populations. At the same time, it can make the individuals of the initial population uniformly dispersed in the search area while enriching the population's variety and minimizing the likelihood of the algorithm becoming stuck in a suboptimal solution. This research uses a Piecewise map to enhance the population initialization of HBA algorithm (Gandomi et al., 2013). The piecewise map can be formulated as follows:

$$x_{k+1} = \begin{cases} x_k/M & 0 \leq x_k < \mathrm{M} \\ (x_k - M)/(0.5 - M) & M \leq x_k < 0.5 \\ (1 - M - x_k)/(0.5 - M) & 0.5 \leq x_k < 1 - M \\ (1 - x_k)/(M) & 1 - M \leq x_k < 1 \end{cases} \tag{30}$$

Where M and x are parameters between 0 and 1.

4.2.2. Segmented optimal decreasing Levy flight strategy

The Levy flight is a stochastic search method that adheres to the Levy distribution, characterized by long-distance leaps. It enhances the global search capabilities of an algorithm while safeguarding against convergence to local optimal. The Levy flight method can be written mathematically as:

$$Lévy = 0.01\mu|\nu|^{-\beta^{-1}} \tag{31}$$

$$\mu \sim N\left(0, \sigma_\mu^2\right), \nu \sim N\left(0, \sigma_\nu^2\right) \tag{32}$$

$$\sigma_\mu = \left[\left(\Gamma(1+\beta)\sin\left(\frac{\pi\beta}{2}\right)\right)\Big/\left(\Gamma\left(\frac{1+\beta}{2}\right)\beta \cdot 2^{\frac{\beta-1}{2}}\right)\right]^{1/\beta}, \sigma_\nu = 1 \tag{33}$$

Γ is a standard Gamma function.

In the process of improving the HBA algorithm, it was found that the traditional Levy flight approach cannot improve the global search ability of the HBA algorithm and will obtain worse optimization results. Therefore, this paper uses piecewise nonlinear decline parameters to control the distance of The Levy flight is integrated with the concept of nonlinear decline to improve the algorithm's capacity to explore the vicinity of the ideal individual. The definition of the segmented optimal decreasing levy flight is as follows:

$$S_{Lévy} = \begin{cases} \delta_1 \times \omega_1 \oplus Lévy, & 0 \leq t \leq t_{\max}/3 \\ \delta_2 \times \omega_2 \oplus Lévy, & t_{\max}/3 < t \leq (2/3)t_{\max} \\ \delta_3 \times \omega_3 \oplus Lévy, & (2/3)t_{\max} < t \leq t_{\max} \end{cases} \tag{34}$$

$S_{Lévy}$ is the segmented optimal Levy flight parameters, δ_1 , δ_2 , δ_3 are the three neighborhood parameters. ω_1, ω_2, and ω_3 are the nonlinear decline coefficients. In the early phase of the iteration, the focus should be on the overall exploration, and the decline coefficient should be used to reduce slowly. In the middle of the iteration, to make the exploration stage to the development stage stably transition, the linear decreasing decline coefficient can ensure that the process proceeds uniformly. In the later stage of iteration, we should focus on local development. Using a decreasing coefficient with a rapid decline can focus on the small-scale optimization process and accelerate convergence. Mathematically, it can be defined as:

$$\begin{cases} \omega_1 = \omega_{\max} - (\omega_{\max} - \omega_{\min}) \times (t/t_{\max})^2 \\ \omega_2 = \omega_{\max} - (\omega_{\max} - \omega_{\min}) \times t/t_{\max} \\ \omega_3 = \omega_{\max} - (\omega_{\max} - \omega_{\min}) \times \left[2t\Big/t_{\max} - (t/t_{\max})^2\right] \end{cases} \tag{35}$$

Where the maximum value is denoted by $\omega_{\max}$, whereas the minimum value is denoted by $\omega_{\min}$.

The segmented optimal decreasing Levy flight strategy is used in the digging and honey phase to enhance the HBA algorithm. Eq. (36) and Eq. (37) replace the original optimal position in the Eq. (28). The equation governing the digging phase is modified in the following manner:

$$x_{new} = x_{prey} + S_{Lévy} + F \times \beta \times I \times x_{prey} + \\ F \times r_3 \times \alpha \times d_i \times |\cos(2pr_4)| \times [1 - \cos(2\pi r_5)] \tag{36}$$

The formula of the honey phase is updated as follows:

$$x_{new} = x_{prey} + S_{Lévy} + F \times r_7 \times \alpha \times d_i \tag{37}$$

4.3. Performance analysis of the IHBA

The use of multiple types of test functions can better reflect the algorithm's optimization capacity, so the study uses two different types of test functions. $f_1(x)$ is a single-peak function, characterized by having just one ideal value. It is mostly employed to assess the search accuracy and convergence speed of an algorithm. $f_2(x)$, $f_3(x)$, $f_4(x)$,and $f_5(x)$ are multimodal functions, which contain multiple local optimal values, which can effectively evaluate the algorithm's capacity to resist slipping into local optimal solutions. The test function information is shown in Table 1.

IHBA was compared with HBA, GWO (Sharma et al., 2016), Harris Hawks Optimization Algorithm (HHO) (Heidari et al., 2019), Manta Ray Foraging Optimization Algorithm (MRFO) (Zhao et al., 2020), and RIME Algorithm (RIME) (Su et al., 2023) to verify the performance of IHBA algorithm under the same number of iterations and runs. The iteration count is set to 1000, and the population size is 50. The algorithm's parameters are displayed in Table 2.

In addition, the five benchmark functions $f_1(x), f_2(x), f_3(x), f_4(x)$ and $f_5(x)$ were independently run 30 times under the same conditions using six algorithms to reduce the randomness of the results. The average convergence curves under different function conditions are shown later in Fig. 2 and Fig. 3. The best value, mean, standard deviation, and average run time of the test results are shown in Table 3.

Fig. 2 and Fig. 3 illustrate the evaluation results of the six methods in test functions f1, f2, f3, f4 and f5, the IHBA approach outperforms the other five algorithms in terms of optimization accuracy and time.

Table 3 illustrates the results of the six algorithms under the unimodal and multi-peak test functions. In the unimodal function, the IHBA algorithm has more advantages in the best value, mean, and standard deviation than GWO, MRFO, and RIME, and lags behind the HHO algorithm in the mean and standard deviation. However, the IHBA algorithm has more advantages than HHO in the average running time of the algorithm. Among the multi-peak functions, the IHBA algorithm's optimization capability is more noticeable, and the average value of the three benchmark functions of $f_3(x)$, $f_4(x)$, $f_5(x)$ is the smallest. The standard deviation significantly differs from other algorithms, showing stronger local search ability. The chaotic initialization strategy and the segmented optimal decreasing levy flight position update method are used to enhance the optimization capacity of the HBA, and better results can be obtained in the best value, mean value, and standard deviation.

Khajuria et al. suggested an improved HBA algorithm (Khajuria et al., 2024). The IHBA algorithm produced average values of 0.1355, −11955.0419, 3.0749e-04, −3.3220, −10.1532, while the five test functions ran under the same conditions and yielded the following average values: 24.3154, −7746.6386, 4.4222e-03, −3.2841, −9.1502, respectively. The IHBA method has superior optimization capabilities compared to the upgraded algorithm proposed by Khajuria. It is suitable for analyzing the economic and environmental dispatch of electricity-hydrogen hybrid microgrids.

Table 1
Test function.

Function	Range	Dim	Theoretical minimum
$f_1(x) = \sum_{i=1}^{29}[100(x_{i+1} - {x_i}^2)^2 + (x_i - 1)^2]$	[-100,100]	30	0
$f_2(x) = -\sum_{i=1}^{30}(x_i \sin(\sqrt{\lvert x_i \rvert}))$	[-500,500]	30	-12569.5
$f_3(x) = \sum_{i=1}^{11}\left[a_i - \frac{x_1(b_i^2 + b_i x_2)}{b_i^2 + b_i x_3 + x_4}\right]^2$	[-5,5]	4	0.0003075
$f_4(x) = -\sum_{i=1}^{4} a_i \exp[-\sum_{j=1}^{n} b_{ij}(x_i - q_{ij})^2]$	[0,1]	6	-3.32199
$f_5(x) = -\sum_{i=1}^{5}[(x - a_i)(x - a_i)^T + c_i]^{-1}$	[0,10]	4	-10.15319

Table 2
Algorithm parameters settings.

Algorithm	Parameters
IHBA	$\delta_1 = 0.02, \delta_2 = 0.02, \delta_3 = 0.06, \beta = 0.5, C = 3$
HBA	$\beta = 6, C = 2$
GWO	a decreases linearly from 2 to 0
HHO	$\beta = 1.5$
MRFO	$S = 2$
RIME	$\omega = 5$

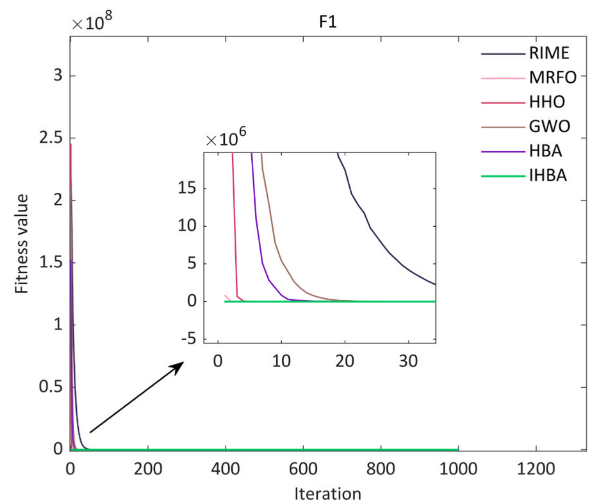

Fig. 2. Algorithm convergence curve of single-peak functions.

4.4. Sensitivity analysis

The IHBA algorithm relies on two key parameters: β and C, which significantly impact its performance. To select the appropriate parameters for optimizing the EED problem, a sensitivity analysis was conducted. The results are shown in Table 4. As the value of C increases, the optimal value initially decreases and then increases, reaching a minimum near $C = 3$. However, the variation between the optimal value and β is more complex. At $\beta < 1$, the optimal value is small. When $C = 3$, the optimal value increases with β and remains stable after $\beta > 1$. Based on the coupling relationship between these two parameters, the chosen values are $C = 3$ and $\beta = 0.5$.

5. Case analysis

5.1. Basic parameters

A recirculating water aquaculture system at a farm in southern China was selected for this study to simulate operations based on the EED model described above. In addition, the heat load is primarily used to maintain the temperature of the cultured water. The simulation was conducted over a 24-hour period, with a step interval of 1 hour, to

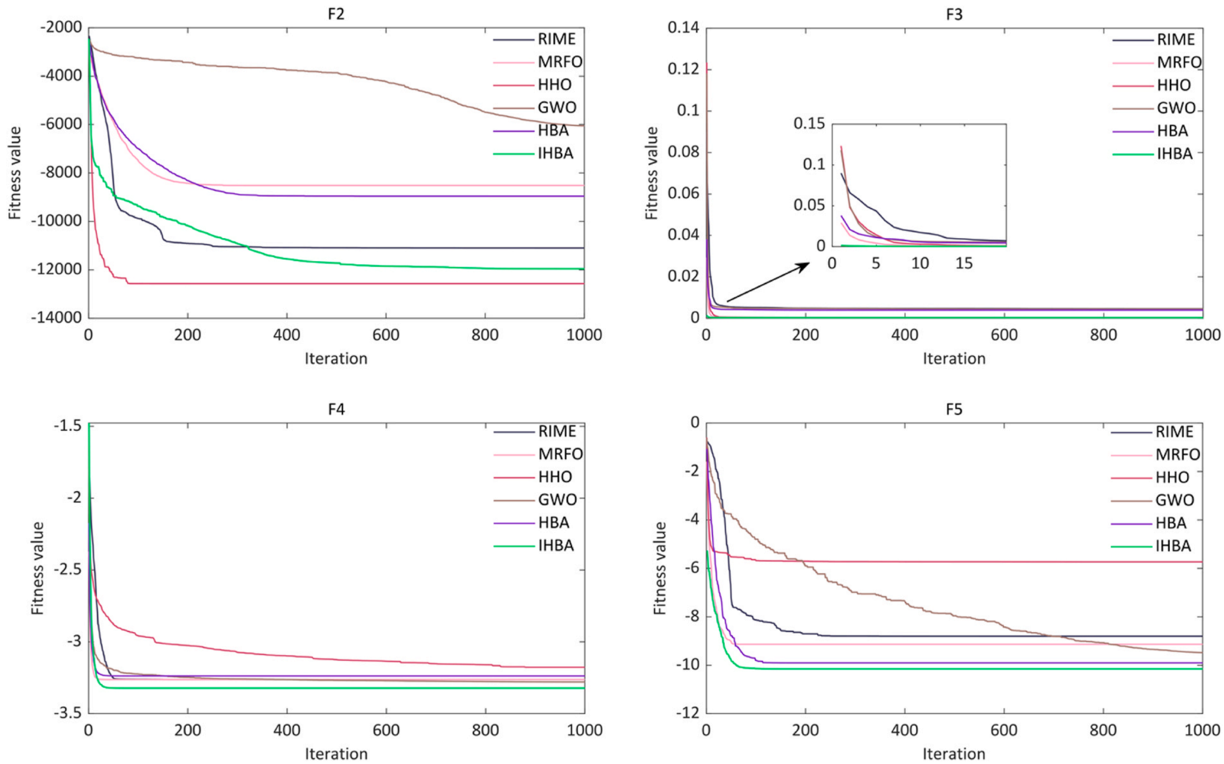

Fig. 3. Algorithm convergence curve of multi-peak functions.

Table 3
Test results.

Function	Algorithms	Best	mean	Std	t_{mean}
$f_1(x)$	IHBA	4.6704e-06	0.1355	0.1382	0.1996
	HBA	19.2443	20.2785	0.6084	0.1922
	GWO	25.2354	26.5046	0.6077	0.1595
	HHO	1.2876e-04	1.8576e-03	2.0941e-03	0.2428
	MRFO	16.5679	17.6496	0.5803	0.2352
	RIME	30.8833	419.5867	628.1628	0.1328
$f_2(x)$	IHBA	-12569.4855	-11955.0419	919.5486	0.1915
	HBA	-10806.5802	-8958.3540	1280.4365	0.1837
	GWO	-8166.7529	-6051.7445	920.6889	0.1585
	HHO	-12569.4866	-12569.4029	0.1821	0.2310
	MRFO	-10003.1378	-8513.4315	730.4762	0.2270
	RIME	-11877.9727	-11097.8016	403.2451	0.1249
$f_3(x)$	IHBA	3.0749e-04	3.0749e-04	2.0405e-09	0.1020
	HBA	3.0749e-04	3.9530e-03	7.8876e-03	0.0998
	GWO	3.0749e-04	4.5060e-03	8.0722e-03	0.0521
	HHO	3.0774e-04	3.2538e-04	2.5429e-05	0.1265
	MRFO	3.0749e-04	4.0295e-04	2.7890e-04	0.1332
	RIME	3.0750e-04	4.6075e-03	8.0174e-03	0.0546
$f_4(x)$	IHBA	-3.3220	-3.3220	4.4574e-05	0.1161
	HBA	-3.3220	-3.2376	0.0904	0.1142
	GWO	-3.3220	-3.2792	0.0627	0.0653
	HHO	-3.3165	-3.1767	0.0808	0.1510
	MRFO	-3.3220	-3.2625	0.0605	0.1526
	RIME	-3.3220	-3.2625	0.0605	0.0657
$f_5(x)$	IHBA	-10.1532	-10.1532	9.5406e-05	0.2196
	HBA	-10.1532	-9.9024	1.3735	0.2201
	GWO	-10.1531	-9.4777	1.7507	0.1324
	HHO	-10.1506	-5.7315	1.7542	0.3310
	MRFO	-10.1532	-9.1336	2.0741	0.3290
	RIME	-10.1532	-8.8028	2.2776	0.1399

optimize the output of the devices within the microgrid. Fig. 4 presents the output data for the wind turbine and PV systems, as well as the demand data for the thermal and electrical loads. ELoad refers to the electricity demand load, while HLoad refers to the heat demand load in Fig. 4.

The specifications of the units used are shown in Table 5.

The time-of-use electricity prices of the day-ahead market are shown in Appendix Table A.1. The emission parameters of power generation units are shown in Appendix Table A.2. The coefficients of pollutant treatment price are shown in Appendix Table A.3. The power conversion

Table 4
Sensitivity analysis for parameters β and C.

Parameters	β=0.1	β=0.5	β=1	β=2	β=3	β=4
C=0.5	21665	22539	20505	20311	20795	21120
C=1	9257	8586	9113	8747	8654	8662
C=1.5	7477	7361	7415	7372	7418	7472
C=2	7044	7068	7065	7102	7146	7104
C=2.5	6998	7075	7381	7018	7122	7080
C=3	6948	6950	6974	7085	7043	7089
C=3.5	6942	7034	7076	7055	7126	7143
C=4	6936	6966	7037	7170	7267	7320
C=5	7048	7294	7401	7492	7679	7775
C=6	7264	7529	7729	7960	8167	8225
C=7	7400	7827	7956	8273	8388	8631

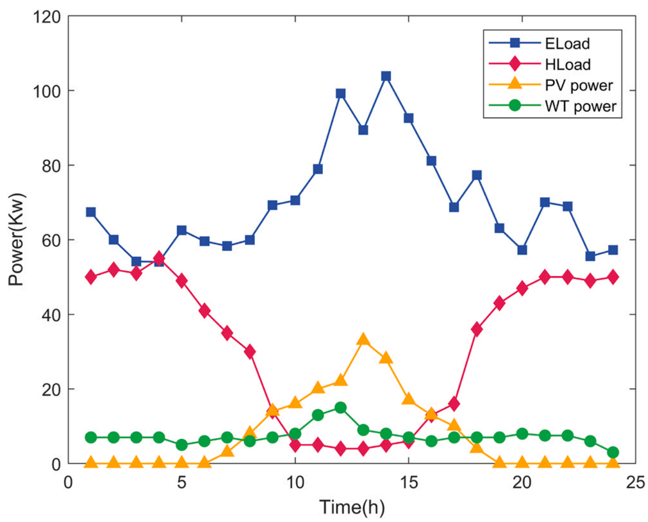

Fig. 4. Power curve of renewable energy generation and load demand.

Table 5
Facilities specifications.

Facilities	Minimum capacity(kW)	Maximum capacity(kW)	Operation and maintenance (RMB/kW)
PV	0	40	4.80
WT	0	20	5.78
ESS	-40	40	2.10
ELE	3	40	3.75
FC	3	40	2.95
HT	0	100	1.04
MT	3	40	4.19
E2H	0	50	-
Grid	-40	40	-

coefficients of facilities are shown in Appendix Table A.4 (Wang et al., 2023a).

5.2. Typical daily power and thermal system analysis

The microgrid described in this paper primarily comprises wind turbine, photovoltaic, battery energy storage system, hydrogen energy storage system, a microturbine, and an electric heating unit, all connected to the external grid. Thermal energy is mainly supplied by the electrolyzer and fuel cell in the HESS, the microturbine, and the electric heating device. In order to investigate the role of HESS participation in the microgrid EED problem, two simulation scenarios are created as described below. The combined economic and environmental costs for the two scenarios are shown in Table 6.

Table 6
Simulation results for two scenarios.

Scenario	The combined economic and environmental costs/RMB
Scenario 1	8332.65
Scenario2	6960.11

(1) Scenario 1: HESS does not participate in the scheduling of the microgrid, and the heat demand is supplied by the E2H and the waste heat of MT;
(2) Scenario 2: HESS participates in the scheduling of the microgrid.

As shown in Table 6, the combined economic and environmental cost of the microgrid with HESS participation in Scenario 2 is 16.47 % lower than that of the conventional microgrid without hydrogen in Scenario 1. Therefore, hydrogen microgrids play a significant role in reducing costs and pollutant emissions. And Fig. 5, Fig. 6, and Fig. 7 illustrate the electrical and thermal optimization results, as well as the renewable energy utilization levels for Scenario 2, respectively.

Fig. 5 shows the power output and electrical load required for the dispatch cycle of the test system. The electrical load (Eload) is determined by combining the electrical load demand shown in Fig. 4 with the load demand of the electric heating unit. In this figure, a negative value for BESS indicates charging, while a positive value indicates discharging. For HESS, a positive value means the fuel cell is in operation, and a negative value means the electrolyzer is in operation.

From 0–8 hours, grid electricity prices are lowest, so the microgrid purchases more grid electricity for power supply, hydrogen production, and battery charging. From 10–15 hours, grid electricity prices are high, but with significant PV and wind power generation, less power is purchased from the grid. From 18–24 hours, grid electricity prices remain high, PV and wind power generation is low, and minimal power is purchased from the grid. During this period, power is primarily supplied by the FC, MT, and BESS. Due to their operating costs, the BESS charge when electricity prices are low and supply supplementary power when prices are high. Throughout the cycle, the system output satisfies power balance constraints. Additionally, from 10 to 15 hours, HESS and MT outputs are minimal, which is further explained in Fig. 6.

The thermal demand of the microgrid is met by the HESS, the MT, and the E2H. During the hours of 0–9 and 16–24, when thermal demand is high, the outputs of HESS and MT are also high, with any shortfall supplemented by the E2H. From 10–15 hours, the thermal demand is low, and the outputs of HESS and MT are sufficient to meet this demand without further increase. The pollutant emissions from HESS are almost zero, while the emissions from MT are high, constrained by environmental objectives. Therefore, the output of HESS is generally greater

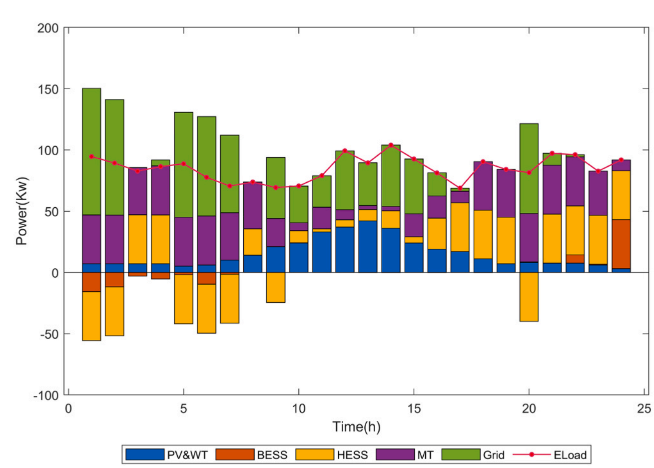

Fig. 5. Daily electric power dispatch results.

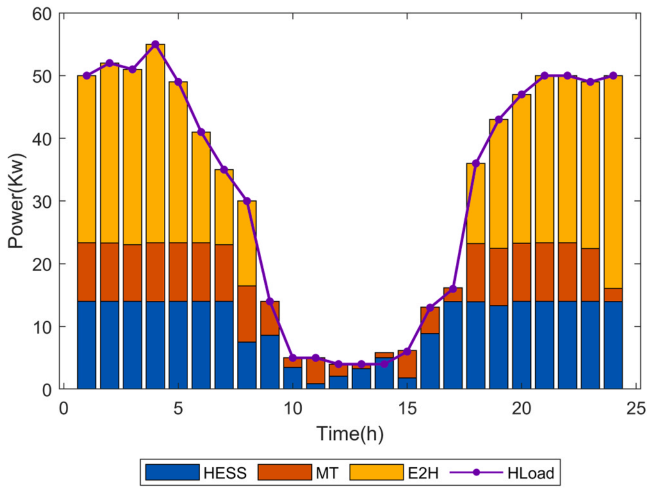

Fig. 6. Daily heat power dispatch results.

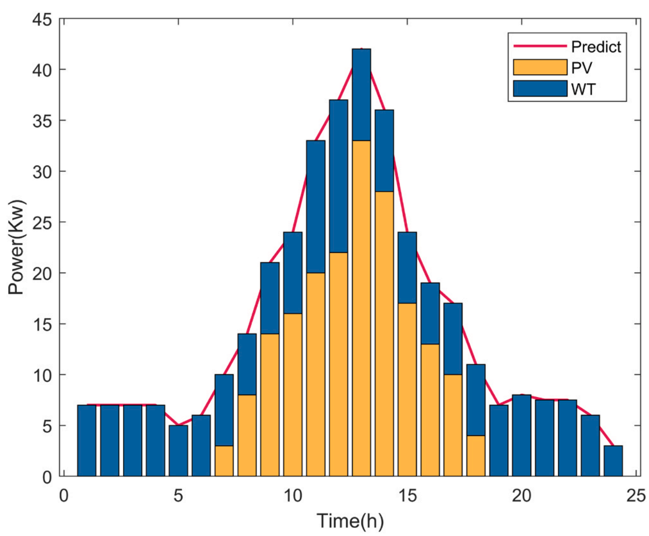

Fig. 7. The sum of renewable energy output and the sum of prediction.

than that of MT to minimize environmental impact.

However, at 14 hours, the system's thermal energy output exceeds the thermal energy demand, leading to some wasted thermal energy. This highlights a shortcoming of the algorithm used in this study, as constraints are enforced using a penalty function, resulting in occasional limit exceedances.

Fig. 7 illustrates the total converted output power from renewable energy sources and the predicted power from these sources throughout the dispatch cycle. During this cycle, priority is given to utilizing power generated by the PV and wind turbine (WT) systems. The utilization rate of renewable energy output is 100 %.

5.3. Performance comparison framework based on TOPSIS in algorithms

This study employs a TOPSIS-based framework to rank the performance of various algorithms and compare their efficiency in solving the electric-hydrogen hybrid microgrid dispatch problem. The comparison framework includes six metrics: best solution, mean solution, standard deviation of the solution, best run time, mean run time, and standard deviation of run time. Twenty independent trials were conducted in the same environment. The best value reflects the optimization capability of the algorithms, while the average indicator shows their accuracy. The standard std indicator measures the precision of the solutions by evaluating how consistently they match the best solution across the 20 trials.

For an equitable comparison, all examined algorithms were subjected to the same maximum number of iterations and quantity of search agents, specifically 1000 iterations and 50 agents. The convergence curves of the IHBA algorithm and five additional techniques are shown in Fig. 8.

Fig. 8 shows that the convergence impact in the later phase is strongly connected to the development of early capability. The data presented in the image demonstrates that the IHBA algorithm exhibits a higher rate of convergence compared to other algorithms during the initial iteration. Moreover, it changes quickly and slower than GWO algorithms but faster than RIME, HHO, MRFO, HBA. However, the convergence speed shifts throughout in the middle iteration of 200–300 generations, and the convergence speed of the IHBA algorithm becomes almost consistent with that of the GWO algorithm. Table 7 and Table 8 show optimization goals and runtime metrics.

From the mean solution in Table 7, the IHBA algorithm optimizes the economic environment of the test system, resulting in a total cost of 6960.11 yuan. This is 5.5 %, 7.9 %, 11.3 %, 2.2 %, and 8.5 % lower than the costs achieved by the HBA, GWO, HHO, MRFO, and RIME algorithms, respectively. Additionally, the standard deviation (std) of the solution results indicates that the IHBA algorithm has high stability. However, ranking different algorithms based on the best solution and the average ideal solution yields very different results. For instance, the optimal and average optimal solutions of the GWO and HHO algorithms differ significantly, making it unreasonable to rank indicators based on a single result.

Table 8 shows that the IHBA method and the HBA algorithm have nearly identical running times, though neither is the most advantageous among the six algorithms. The MRFO method produces optimization results comparable to those of the IHBA algorithm but has one of the slowest running times, only faster than the GWO. The correlation between time and optimal value may exhibit an inverse relationship. When choosing algorithms, it's essential to consider the problem's characteristics along with the requirements for result accuracy and computational efficiency. Therefore, this study establishes a comparison framework of meta-heuristic algorithms based on TOPSIS, which comprehensively considers the optimal solution and the related indicators of running time. The comparison results of the six algorithms are shown in Table 9 below.

In Table 9, the value C_i of the IHBA algorithm is 0.26. This proximity to the ideal algorithm for solving the electric-hydrogen hybrid microgrid scheduling problem is significantly higher than that of the original HBA

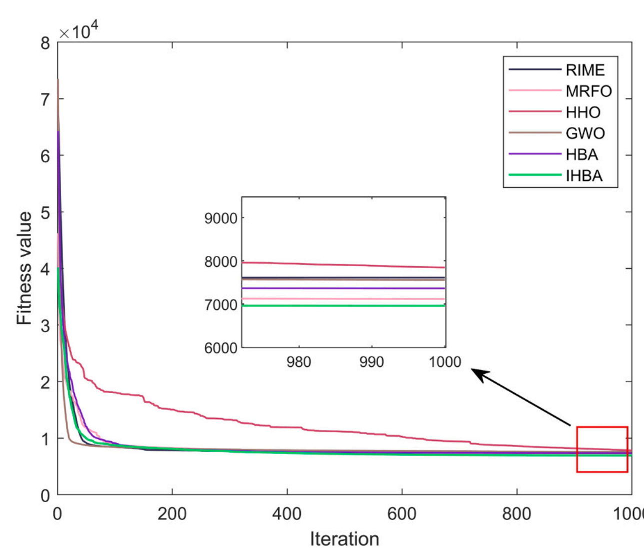

Fig. 8. The average convergence curves.

Table 7
Optimization and comparison results of different algorithms.

Algorithms	Best Value	Mean value	Std
IHBA	6693.01	6960.11	144.96
HBA	7035.58	7363.08	195.45
GWO	7252.68	7557.32	224.60
HHO	7514.94	7847.20	291.32
MRFO	6878.48	7115.75	135.76
RIME	7285.20	7609.88	161.95

Table 8
Runtime results of different algorithms.

Algorithms	t_{best}	t_{mean}	Std
IHBA	1.47	1.60	0.045
HBA	1.42	1.58	0.048
GWO	1.38	1.43	0.044
HHO	2.75	2.94	0.087
MRFO	2.47	2.53	0.047
RIME	1.31	1.36	0.032

Table 9
Comparison results based on the framework.

Algorithms	d_i^+	d_i^-	C_i	rank
IHBA	0.11	0.92	0.26	1
HBA	0.32	0.73	0.21	2
GWO	0.47	0.70	0.18	4
HHO	1	0	0	6
MRFO	0.51	0.66	0.17	5
RIME	0.46	0.77	0.19	3

algorithm. For instance, considering the MRFO algorithm, although it achieves a smaller optimal value, ranking second, its running time is not dominant, leading to a fifth-place ranking under the comparison framework. As previously mentioned, the IHBA algorithm exhibits the best performance in balancing optimization capability and efficiency. Moreover, the comparison framework for different meta-heuristic algorithms demonstrates high effectiveness, enabling a fair selection of the best algorithm for solving the EED problem.

6. Conclusion

To enhance the efficiency and environmental sustainability of microgrids while meeting power and heat demands, this paper introduces a novel EED model for hydrogen-containing microgrids utilizing waste heat. Additionally, an improved version of the IHBA algorithm and a comparison framework are proposed to optimize the EED problem. The key conclusions of this study are as follows:

(1) The developed EED model for hydrogen-energy-containing microgrids significantly enhances the economic and environmental performance of microgrids. By reducing the reliance on MT and decreasing pollution emissions, the system optimally utilizes peak and off-peak electricity prices. The ESS efficiently stores energy during low-price periods and supplies electricity during high-price periods. Simultaneously, the HESS produces and sells excess hydrogen, leading to a 16.47 % reduction in comprehensive economic and environmental costs compared to traditional microgrids without HESS.

(2) The proposed model effectively utilizes waste heat from the HESS and MT without requiring a heat storage device, thereby minimizing thermal energy wastage and enhancing energy utilization while meeting the system's thermal energy demand.

(3) In the algorithm comparison framework, the IHBA algorithm demonstrates superior performance, offering the best combination of optimization capability and efficiency for solving the hydrogen-containing microgrid EED problem. The combined EED cost of the IHBA optimized microgrid is reduced by 5.5 %, 7.9 %, 11.3 %, 2.2 %, and 8.5 % compared to the models optimized using the HBA, GWO, HHO, MRFO, and RIME algorithms, respectively.

(4) This method effectively addresses the EED problem by incorporating hydrogen energy, thereby reducing the operating cost and pollution emissions of the MG system. However, the EED model employs fewer constraints, which may not be sufficient for complex scenarios. Additionally, the improved algorithm proposed in this paper is limited to day-ahead scheduling optimization and is not applied to multi-time scale optimization. The research presented in this paper has certain limitations. Future work could include adding more constraints and exploring the application of the improved method for annual cost optimization in microgrids.

CRediT authorship contribution statement

Chen Yang: Writing – review & editing, Methodology, Conceptualization. **Jingxiang Xu:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Guangzhe Jin:** Writing – original draft, Software, Methodology, Conceptualization. **Kaixin Huang:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Table A.1
The parameters of time-of-use price

Power purchase price, RMB/kWh	Power sale price, RMB/kWh	Time interval
1.350	1.040	9:00–12:00 19:00–24:00
0.900	0.693	8:00–9:00 12:00–19:00
0.500	0.385	0:00–8:00

Table A.2
The coefficients of pollutant emission

Types of micro-source	Pollutant (g/kWh)			
	CO	CO_2	SO_2	NO_x
MT	0.047	724	0.0036	0.2
Grid	0.081	889	0.8	0.6

Table A.3
The coefficients of pollutant treatment price

Pollutant	Remediation cost (RMB/kg)
CO	10.29
CO_2	0.21
SO_2	14.842
NO_x	62.964

Table A.4
The coefficients of power conversion

Facilities	Coefficients
ELE	$\eta_{ELE\_H} = 0.6, \eta_{ELE\_Q} = 0.88$
FC	$\eta_{FC\_E} = 0.6, \eta_{FC\_Q} = 0.88$
MT	$\eta_{MT\_E} = 0.632, \eta_{MT\_Q} = 0.234$
E2H	$\eta_{E2H\_Q} = 0.98$

References

Chen, S., Kharrazi, A., Liang, S., Fath, B.D., Lenzen, M., Yan, J., 2020. Advanced approaches and applications of energy footprints toward the promotion of global sustainability. Appl. Energy 261, 114415.

Emrani, A., Berrada, A., Bakhouya, M., 2022. Optimal sizing and deployment of gravity energy storage system in hybrid PV-Wind power plant. Renew. Energy 183, 12–27.

Fonseca, J.D., Camargo, M., Commenge, J.-M., Falk, L., Gil, I.D., 2019. Trends in design of distributed energy systems using hydrogen as energy vector: a systematic literature review. Int. J. Hydrog. Energy 44, 9486–9504.

Gandomi, A.H., Yang, X.S., Talatahari, S., Alavi, A.H., 2013. Firefly algorithm with chaos. Commun. Nonlinear Sci. Numer. Simul. 18, 89–98.

Ghenai, C., Salameh, T., Merabet, A., 2020. Technico-economic analysis of off grid solar PV/Fuel cell energy system for residential community in desert region. Int. J. Hydrog. Energy 45, 11460–11470.

Hashim, F.A., Houssein, E.H., Hussain, K., Mabrouk, M.S., Al-Atabany, W., 2022. Honey badger algorithm: new metaheuristic algorithm for solving optimization problems. Math. Comput. Simul. 192, 84–110.

Heidari, A.A., Mirjalili, S., Faris, H., Aljarah, I., Mafarja, M., Chen, H., 2019. Harris hawks optimization: algorithm and applications. Future Gener. Comput. Syst. 97, 849–872.

Ju, L., Liu, L., Han, Y., Yang, S., Li, G., Lu, X., Liu, Y., Qiao, H., 2023. Robust Multi-objective optimal dispatching model for a novel island micro energy grid incorporating biomass waste energy conversion system, desalination and power-to-hydrogen devices. Appl. Energy 343, 121176.

K. V, M. V, Guerrero, J.M., Bazmohammadi, N., 2022. Energy management system using Mimosa pudica optimization technique for microgrid applications. Energy 244, 122605.

Khajuria, R., Yelisetti, S., Lamba, R., Kumar, R., 2024. Optimal model parameter estimation and performance analysis of PEM electrolyzer using modified honey badger algorithm. Int. J. Hydrog. Energy 49, 238–259.

Li, L.-L., Zheng, S.-J., Tseng, M.-L., Liu, Y.-W., 2021. Performance assessment of combined cooling, heating and power system operation strategy based on multi-objective seagull optimization algorithm. Energy Convers. Manag. 244, 114443.

Liu, G., Yuan, J., Lin, K.-P., Miao, Y., Li, R., 2024. Developing an improved chameleon swarm algorithm for combined cooling, heating and power micro-grid system. Expert Syst. Appl. 237, 121540.

Mirhoseini, P., Ghaffarzadeh, N., 2020. Economic battery sizing and power dispatch in a grid-connected charging station using convex method. J. Energy Storage 31, 101651.

Mohseni, S., Brent, A.C., 2020. Economic viability assessment of sustainable hydrogen production, storage, and utilisation technologies integrated into on- and off-grid micro-grids: a performance comparison of different meta-heuristics. Int. J. Hydrog. Energy 45, 34412–34436.

Nikmehr, N., Ravadanegh, S.N., 2015. Optimal PoWer Dispatch of Multi-microgrids at Future Smart Distribution Grids. IEEE Trans. Smart Grid 6, 1648–1657.

Roy, A., Auger, F., Dupriez-Robin, F., Bourguet, S., Tran, Q.T., 2020. A multi-level demand-side management algorithm for offgrid multi-source systems. Energy 191, 116536.

Ruiming, F., 2019. Multi-objective optimized operation of integrated energy system with hydrogen storage. Int. J. Hydrog. Energy 44, 29409–29417.

Sharma, S., Bhattacharjee, S., Bhattacharya, A., 2016. Grey wolf optimisation for optimal sizing of battery energy storage device to minimise operation cost of microgrid. Iet Gener. Transm. Distrib. 10, 625–637.

Shuai, H., Fang, J., Ai, X., Tang, Y., Wen, J., He, H., 2019. Stochastic optimization of economic dispatch for microgrid based on approximate dynamic programming. IEEE Trans. Smart Grid 10, 2440–2452.

Sohani, A., Delfani, F., Fassadi Chimeh, A., Hoseinzadeh, S., Panchal, H., 2022. A conceptual optimum design for a high-efficiency solar-assisted desalination system based on economic, exergy, energy, and environmental (4E) criteria. Sustain. Energy Technol. Assess. 52, 102053.

Su, H., Zhao, D., Heidari, A.A., Liu, L., Zhang, X., Mafarja, M., Chen, H., 2023. RIME: a physics-based optimization. Neurocomputing 532, 183–214.

Vivas, F.J., De las Heras, A., Segura, F., Andujar, J.M., 2018. A review of energy management strategies for renewable hybrid energy systems with hydrogen backup. Renew. Sustain. Energy Rev. 82, 126–155.

W.G.M. D.H. Wolpert, 1997. No free lunch theorems for optimization. IEEE Trans. Evolut. Comput. 1, 67–82.

Wang, L., An, X., Xu, H., Zhang, Y., 2022. Multi-agent-based collaborative regulation optimization for microgrid economic dispatch under a time-based price mechanism. Electr. Power Syst. Res. 213, 108760.

Wang, J., Chen, B., Che, Y., 2023b. Bi-level sizing optimization of a distributed solar hybrid CCHP system considering economic, energy, and environmental objectives. Int. J. Electr. Power Energy Syst. 145, 108684.

Wang, P., Liu, D., Mukherjee, A., Agrawal, M., Zhang, H., Agathokleous, E., Qiao, X., Xu, X., Chen, Y., Wu, T., Zhu, M., Saikawa, E., Agrawal, S.B., Feng, Z., 2023a. Air pollution governance in China and India: comparison and implications. Environ. Sci. Policy 142, 112–120.

Wang, Y., Qin, Y., Ma, Z., Wang, Y., Li, Y., 2023a. Operation optimisation of integrated energy systems based on cooperative game with hydrogen energy storage systems. Int. J. Hydrog. Energy 48, 37335–37354.

Zhang, S., Hu, W., Cao, X., Du, J., Bai, C., Liu, W., Tang, M., Zhan, W., Chen, Z., 2023. Low-carbon economic dispatch strategy for interconnected multi-energy microgrids considering carbon emission accounting and profit allocation. Sustain. Cities Soc. 99, 104987.

Zhao, W., Zhang, Z., Wang, L., 2020. Manta ray foraging optimization: an effective bio-inspired optimizer for engineering applications. Eng. Appl. Artif. Intell. 87, 103300.

Zheng, Q., Gu, Y., Liu, Y., Ma, J., Peng, M., 2023. Chaotic particle swarm algorithm-based optimal scheduling of integrated energy systems. Electr. Power Syst. Res. 216, 108979.

Zhou, X., Ai, Q., 2019. Distributed economic and environmental dispatch in two kinds of CCHP microgrid clusters. Int. J. Electr. Power Energy Syst. 112, 109–126.